

# Fundamentals of Meteorological Observations

## Unit 2: Conventional Meteorological Instruments

Trainer notes — for internal training use on the Capacity Connect platform.

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### Learning Objectives

- Identify the standard instruments used at a conventional surface observatory
- Explain the working principle of each major instrument
- Recognise common exposure and maintenance issues

### Temperature & Humidity

Dry-bulb and wet-bulb thermometers, mounted in a Stevenson screen, remain the reference instruments for air temperature and, via the psychrometric relationship, relative humidity. Maximum and minimum thermometers record the day's extremes and must be reset correctly at each main observation to avoid carrying over a stale reading.

### Pressure

Mercury barometers (increasingly replaced by precision aneroid and digital barometers) measure station-level pressure, which is then reduced to mean sea level using the station's elevation so that pressure values are comparable across stations at different altitudes. Barograph traces provide a continuous record useful for tracking approaching weather systems.

### Rainfall & Wind

The standard non-recording rain gauge (127 mm diameter funnel) is read once or twice daily and remains the reference against which even automatic rain gauges are periodically checked. Wind is measured using a cup anemometer for speed and a wind vane for direction, mounted at the standard 10 m height over open exposure to avoid distortion from nearby structures.

### Summary

Every automated sensor IMD deploys today is ultimately validated against these conventional instruments, which is why understanding their working principle and correct exposure remains a core observer skill.